

# Time Series Analysis

## Chapter 3 — Stationary time series

Aymen Ben Brik  
aymen.benbrik@esprit.tn  
Esprit School of Business

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# Chapter 1

## Stationary time series

### Introduction

After decomposing the observed series into trend, seasonality and residual (Chapter ??), we usually want the residual to be *stationary* so that the modelling tools of Chapter ?? apply. This chapter formalises stationarity, introduces the autocovariance and autocorrelation functions (ACF and PACF), states the Wold decomposition theorem, and provides *white-noise tests* (Box–Pierce, Ljung–Box) to validate model residuals.

### 1.1 Stationarity

#### 1.1.1 Strict and weak stationarity

##### Definition 1 – Strict (strong) stationarity

A process  $\{X_t\}_{t \in \mathbb{Z}}$  is *strictly stationary* if, for every integer  $k$ , every  $t_1 < \dots < t_k$  and every  $h \in \mathbb{Z}$ ,

$$(X_{t_1}, X_{t_2}, \dots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, X_{t_2+h}, \dots, X_{t_k+h}).$$

##### Definition 2 – Weak (covariance) stationarity

A process  $\{X_t\}$  with  $\mathbb{E}[X_t^2] < \infty$  is *weakly stationary* if there exist  $\mu \in \mathbb{R}$  and a function  $\gamma : \mathbb{Z} \rightarrow \mathbb{R}$  such that

$$\mathbb{E}[X_t] = \mu, \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for all } t, h \in \mathbb{Z}.$$

The function  $\gamma$  is the *autocovariance function* (ACVF).

##### Proposition 3 – Strict $\Rightarrow$ weak (under $L^2$ )

If  $\{X_t\}$  is strictly stationary with  $\mathbb{E}[X_t^2] < \infty$ , then  $\{X_t\}$  is weakly stationary.

*Proof.* By strict stationarity,  $X_t \stackrel{d}{=} X_0$ , so  $\mathbb{E}[X_t] = \mathbb{E}[X_0] = \mu$  does not depend on  $t$ . Similarly,  $(X_t, X_{t+h}) \stackrel{d}{=} (X_0, X_h)$ , hence  $\text{Cov}(X_t, X_{t+h}) = \text{Cov}(X_0, X_h)$  depends only on  $h$ .  $\square$

##### Remark – The converse is false in general

A process can be weakly stationary without being strictly stationary if it has time-varying higher-order moments. For Gaussian processes (only mean and covariance matter), the

two notions coincide.

### 1.1.2 Two canonical examples

#### Definition 4 – White noise

$(\varepsilon_t)$  is a *white noise*  $\text{WN}(0, \sigma^2)$  if  $\mathbb{E}[\varepsilon_t] = 0$ ,  $\text{Var}[\varepsilon_t] = \sigma^2$  and  $\text{Cov}(\varepsilon_t, \varepsilon_{t+h}) = 0$  for  $h \neq 0$ . A *strong white noise* requires the  $\varepsilon_t$  to be i.i.d.; a Gaussian white noise has  $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ .

#### Example 5 – Random walk is *not* stationary

Let  $\varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \text{WN}(0, \sigma^2)$  and  $X_t = X_{t-1} + \varepsilon_t$  with  $X_0 = 0$ . Then  $X_t = \sum_{s=1}^t \varepsilon_s$  and

$$\mathbb{E}[X_t] = 0, \quad \text{Var}[X_t] = t\sigma^2, \quad \text{Cov}(X_t, X_{t+h}) = t\sigma^2 \text{ for } h \geq 0.$$

The variance grows linearly in  $t$ , so  $X_t$  is non-stationary. This is the canonical example used by unit-root tests in Chapter ??.

## 1.2 Autocovariance and autocorrelation

#### Definition 6 – ACVF and ACF

For a weakly stationary process with mean  $\mu$ :

$$\gamma(h) = \text{Cov}(X_t, X_{t+h}), \quad \rho(h) = \frac{\gamma(h)}{\gamma(0)} \in [-1, 1].$$

#### Proposition 7 – Properties of $\gamma$

1.  $\gamma(0) = \text{Var}[X_t] \geq 0$ ;
2.  $|\gamma(h)| \leq \gamma(0)$  (Cauchy–Schwarz);
3.  $\gamma$  is even:  $\gamma(-h) = \gamma(h)$ ;
4.  $\gamma$  is positive semidefinite: for any  $n$  and any  $a_1, \dots, a_n \in \mathbb{R}$ ,  $\sum_{i,j=1}^n a_i a_j \gamma(t_i - t_j) \geq 0$ .

*Proof.* (1) and (3) follow from the definition. (2) follows from Cauchy–Schwarz applied to  $X_t - \mu$  and  $X_{t+h} - \mu$ . (4) is the standard fact that any matrix of the form  $\Sigma_{ij} = \gamma(t_i - t_j)$  is the covariance matrix of  $(X_{t_1}, \dots, X_{t_n})$ , hence positive semidefinite.  $\square$

### 1.2.1 Sample ACF and Bartlett bands

#### Definition 8 – Sample ACVF and ACF

For  $X_1, \dots, X_T$  with sample mean  $\bar{X}$ :

$$\hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (X_t - \bar{X})(X_{t+h} - \bar{X}), \quad \hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

**Theorem 9 – Bartlett’s formula for white noise**

If  $\{X_t\}$  is a white noise with finite fourth moment, then for any fixed  $h \geq 1$ :

$$\sqrt{T} \hat{\rho}(h) \xrightarrow{d} \mathcal{N}(0, 1).$$

The standard test bands  $\pm 1.96/\sqrt{T}$  on the ACF plot are asymptotic 95% confidence intervals *under the white-noise null*.

**Remark**

Bartlett’s formula gives a more general result for non-white processes (involves a sum over the squared autocorrelations); see Brockwell & Davis, *Time Series: Theory and Methods*, Chapter 7.

### 1.3 Partial autocorrelation

**Definition 10 – Partial autocorrelation function**

The PACF  $\alpha(h)$  at lag  $h$  is the correlation between  $X_t$  and  $X_{t+h}$  *after removing the linear effect* of the intermediate observations  $X_{t+1}, \dots, X_{t+h-1}$ :

$$\alpha(h) = \text{Corr}(X_t - \hat{X}_t, X_{t+h} - \hat{X}_{t+h}),$$

where  $\hat{X}_t$  and  $\hat{X}_{t+h}$  are the linear projections of  $X_t$  and  $X_{t+h}$  on the span of  $\{X_{t+1}, \dots, X_{t+h-1}\}$ .

**Remark – Recursive computation**

The PACF is computed by the *Durbin–Levinson* algorithm: it solves the Yule–Walker equations of order  $h$  and reads off the coefficient of  $X_{t+h-h} = X_t$  in the linear regression of  $X_{t+h}$  on  $X_{t+1}, \dots, X_{t+h}$ . Most software (Python `statsmodels`, R `stats`) returns the PACF in  $O(h)$  operations per lag.

### 1.4 ACF / PACF signatures of standard processes

| Process        | $\rho(h)$ behaviour          | $\alpha(h)$ behaviour          |
|----------------|------------------------------|--------------------------------|
| WN             | $\rho(h) = 0$ for $h \neq 0$ | $\alpha(h) = 0$ for $h \geq 1$ |
| AR( $p$ )      | decays (exp. / sinusoidal)   | cuts off at $h = p$            |
| MA( $q$ )      | cuts off at $h = q$          | decays                         |
| ARMA( $p, q$ ) | decays after $h > q$         | decays after $h > p$           |

**Box–Jenkins identification.** Plot the sample ACF and PACF, then use the table above to read off candidate orders  $(p, q)$ .

### 1.4.1 AR(1) and MA(1) — closed-form ACF

#### Example 11 – AR(1) ACF

For  $X_t = \phi X_{t-1} + \varepsilon_t$  with  $\varepsilon_t \sim \text{WN}(0, \sigma^2)$  and  $|\phi| < 1$ :

$$\rho(h) = \phi^{|h|}, \quad h \in \mathbb{Z}, \quad \alpha(1) = \phi, \quad \alpha(h) = 0 \text{ for } h \geq 2.$$

The ACF decays exponentially; the PACF cuts off at lag 1.

#### Example 12 – MA(1) ACF

For  $X_t = \varepsilon_t + \theta \varepsilon_{t-1}$  with  $\varepsilon_t \sim \text{WN}(0, \sigma^2)$ :

$$\rho(0) = 1, \quad \rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(h) = 0 \text{ for } h \geq 2.$$

The ACF cuts off at lag 1; the PACF decays.

*Proof of  $\rho(1) = \theta/(1 + \theta^2)$  in MA(1).*  $\text{Var}[X_t] = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})^2] = (1 + \theta^2)\sigma^2$ . Then  $\text{Cov}(X_t, X_{t+1}) = \mathbb{E}[(\varepsilon_t + \theta \varepsilon_{t-1})(\varepsilon_{t+1} + \theta \varepsilon_t)] = \theta \sigma^2$  since  $\varepsilon_t$ 's are uncorrelated. Hence  $\rho(1) = \theta \sigma^2 / [(1 + \theta^2)\sigma^2] = \theta/(1 + \theta^2)$ .  $\square$

## 1.5 Wold decomposition

#### Theorem 13 – Wold's theorem, 1938

Every *purely non-deterministic* weakly stationary process  $\{X_t\}$  admits a unique representation

$$X_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \psi_0 = 1, \quad \sum_{j=0}^{\infty} \psi_j^2 < \infty,$$

where  $(\varepsilon_t)$  is a white-noise process (the *innovations*) of variance  $\sigma^2 = \text{Var}[\varepsilon_t]$ .

#### Remark – Why it matters

- **Universality:** every stationary process can be viewed as an *infinite MA process*.
- **Justification of ARMA:** ARMA models are finite-dimensional approximations of the Wold representation (with a rational  $\psi(z)$ ).
- **Forecasting:** optimal linear predictors and their prediction errors are obtained from the Wold coefficients.

## 1.6 White-noise tests

After fitting a decomposition or an ARMA model, the residuals  $\hat{\varepsilon}_t$  should look like white noise. Two classical tests check this jointly across several lags.

**Definition 14 – Box–Pierce and Ljung–Box statistics**

For sample ACF  $\hat{\rho}(h)$  of the residuals up to lag  $m$ :

$$Q_{BP} = T \sum_{h=1}^m \hat{\rho}(h)^2, \quad Q_{LB} = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}(h)^2}{T-h}.$$

**Theorem 15 – Asymptotic distribution under  $H_0$** 

Under  $H_0$  : *the residuals are white noise*, both statistics are asymptotically  $\chi^2_{m-p-q}$ , where  $p+q$  is the number of fitted ARMA parameters (zero if no parametric model has been fitted).

**Decision rule.** Reject  $H_0$  at level  $\alpha$  if  $Q > \chi^2_{m-p-q, 1-\alpha}$ , equivalently  $p\text{-value} < \alpha$ .

**Choosing  $m$ .** A few rules of thumb:

- For yearly data:  $m \in \{10, 20\}$ .
- For monthly data:  $m \in \{12, 24, 36\}$  (covers seasonality).
- For high-frequency data:  $m$  proportional to expected autocorrelation length.

**Warning**

The Ljung–Box test is preferred over Box–Pierce in finite samples: its small-sample distribution is closer to the  $\chi^2$  asymptotic limit, especially when  $m/T$  is not negligible.

**Summary****Key takeaways.**

- *Strict stationarity* = invariance of all finite-dimensional distributions; *weak stationarity* = only first two moments. Strict  $\Rightarrow$  weak (under  $L^2$ ); converse holds for Gaussian processes.
- The *autocovariance*  $\gamma(h)$  describes the dependence structure; the *autocorrelation*  $\rho(h)$  is its normalised version.
- ACF and PACF have characteristic patterns that allow *Box–Jenkins identification* of AR(p) / MA(q) / ARMA(p,q).
- *Wold decomposition*: every stationary process is an infinite moving average of innovations.
- *Ljung–Box test* validates that residuals look like white noise.